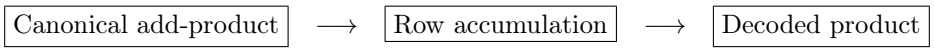
Independent product accumulation review

Reviewer /root/monic_review. 2026-11-01.

Checked result. Fresh isolated replay passed ten main declarations, eight producer controls and six independent controls. All 24 named axiom readbacks use only propext, Classical.choice and Quot.sound. Mathematical fidelity is faithful at the stated model scope; full native execution correspondence remains partial.

Independent derivation. Before reading the new proof bodies, the reviewer read native pmul and wrote the expected completed-row and partial-row convolution invariant. Completed rows contribute a prefix of A times B; partial row i contributes $a_i X^i$ times the processed prefix of B. The producer's offset/remaining-row invariant agrees with this derivation.



Exact premises and boundaries. Prime $0 < p < 2^{31}$ and canonical operands. Each raw product is below 2^{62} and the modular addition below 2^{32}. No monicity, normalization or nonzero operand assumption is needed. Both empty branches precede allocation. Indexed events preserve row-major order, repeated destinations accumulate, and final trimming preserves polynomial value.

Native correspondence. Source pmul lines 835–848 agrees in loop order, update formula, empty branches and final trim. Interpretation is odd prime-field extension degree one with diagnostic tables absent. Mathematical Nat/Fin bounds do not alone prove native usize allocation arithmetic, allocation success, counter/instrumentation behavior or compiled equivalence. Those are distinct obligations.

Independent controls. Asymmetric operands with internal zeros, swapped operand lengths, nonempty zero operand, both empty branches and maximal canonical residue accumulation at modulus 17 all passed kernel evaluation. No timing or scientific trial was performed.

Workflow. Prospective local plan, source-first derivation, separate reviewer, no sibling reports, fresh replay and named axiom checks completed. The plan was not precommitted; inputs were stable but not published immutably at dispatch. Full imported dependency rebuild, independent kernel rechecker, multi-reviewer independence gate and coordinator publication remain separate. Imported helper binaries and cached Mathlib retain their stated trust boundary. No official scientific transition is made.

Detailed findings: REVIEW.md. Structured results: status.json. Commands and output: receipts.json, extra-receipts.json and module logs. Exact input bindings and editable SVG accompany this report. Source-first invariant and attacks are preserved in review-plan.md.